Q1

Choose the correct answer in each of the following:

1) Which component links different object code files into a single executable?
a) Assembler
b) Linker
c) Preprocessor
d) Parser

2) What is the role of semantic analysis in compilation?
a) Token generation
b) Linking object code
c) Generating executable code
d) Type checking and detecting semantic errors

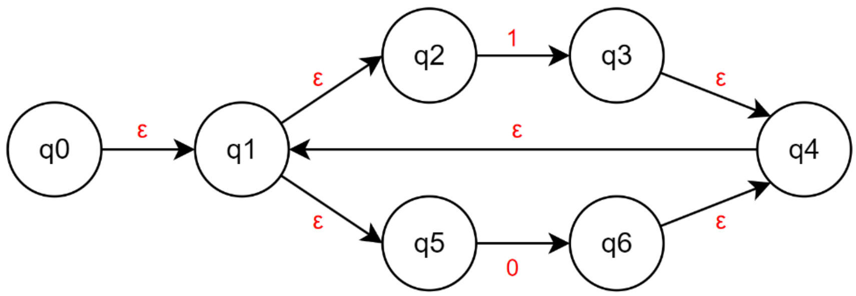
3) Which of the following is TRUE about the final phase of the compiler?
a) Optimizes the intermediate code.
b) Generates object code from optimized intermediate code.
c) Converts object code back to source code.
d) Generates three-address code.

4) What kind of error is detected by the lexical analyzer?
a) Undefined variables
b) Incorrect data types
c) Illegal characters
d) Stack overflow

5) Which compiler phase is responsible for reporting most syntax errors?
a) Semantic Analyzer
b) Code Generator
c) Lexical Analyzer
d) Syntax Analyzer

Q2

a. Convert the following NFA into its equivalent DFA. (Fill the table)



Details:

NFA State	DFA State	0	1
	A		
	B		

b. Using **McNaughton-Yamada-Thompson** algorithm convert the regular expression to NFA:
ab(a|b*c)*a

Q3

Eliminate left recursion (only one **very easy** grammar)

Q4

a. Construct the canonical LR(0) collection for the following grammar using CLOSURE and GOTO. Stop after completing the states which are coming from (I_0)

$$S' \longrightarrow S$$
$$S \longrightarrow AaAb$$
$$S \longrightarrow BbBa$$
$$A \longrightarrow b$$
$$B \longrightarrow a$$

b. Write the final states

Q5

- $E \rightarrow T\ E'$
- $E' \rightarrow +\ T\ E' \mid \varepsilon$
- $T \rightarrow F\ T'$
- $T' \rightarrow *\ F\ T' \mid \varepsilon$
- $F \rightarrow (E) \mid id$

Given the above grammar Show the sequence of procedure calls (and returns) performed by a recursive descent parser for the following:

E:

T':

F:

Q6

$$E \rightarrow E+T \mid T$$
$$T \rightarrow T * F \mid F$$
$$F \rightarrow N \mid E$$
$$N \rightarrow D$$
$$D \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$$

(a) Draw pars tree for: 9 * 5 + 4 * 3 + 2

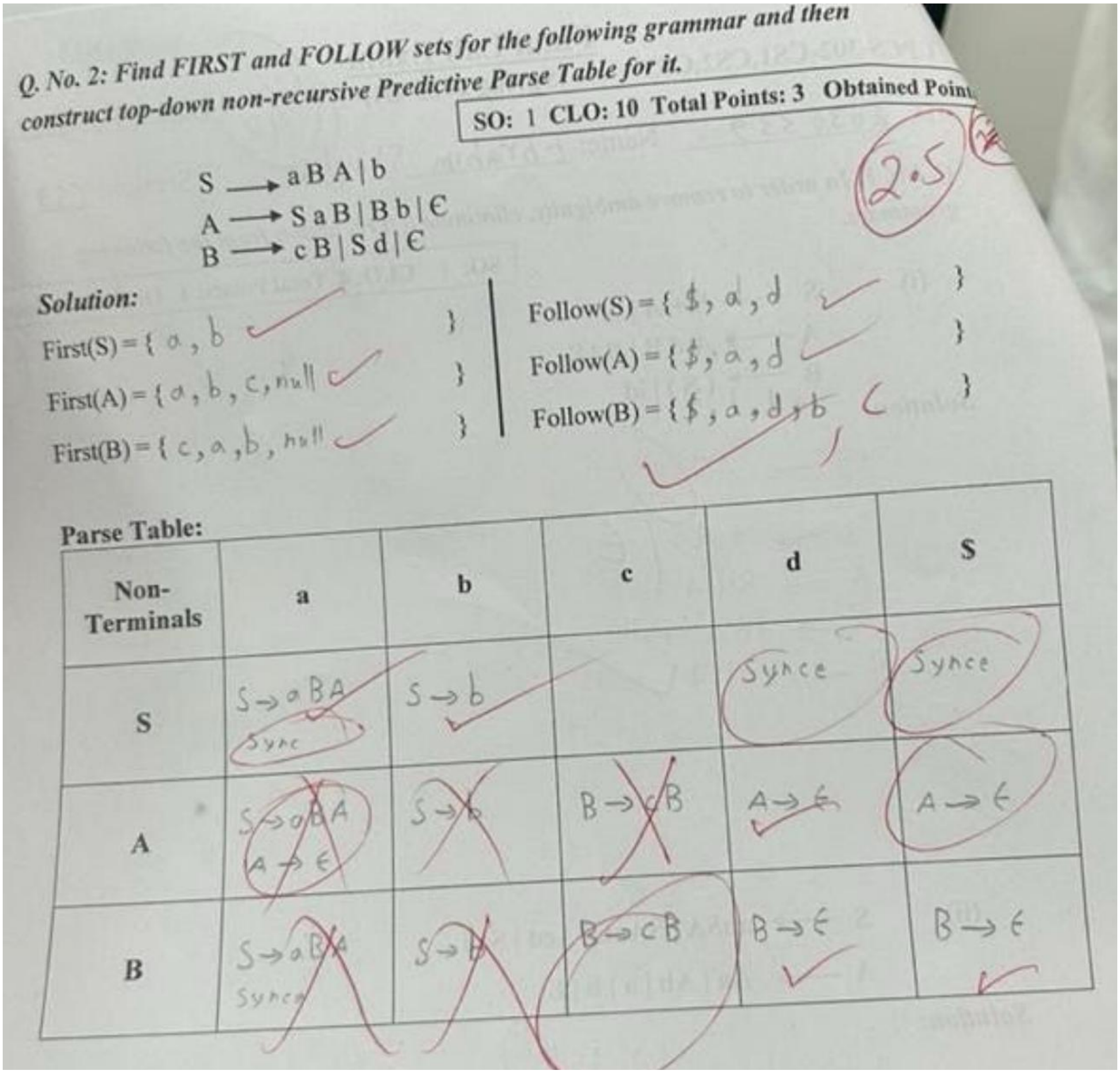
(b) Find the left derivation for: 8 * 5 + 9

(c) is the grammar ambiguous for (*,+)? explain your answer.

Q7

(a)

he gave us different grammar



Q. No. 2: Find FIRST and FOLLOW sets for the following grammar and then construct top-down non-recursive Predictive Parse Table for it.

SO: 1 CLO: 10 Total Points: 3 Obtained Points 2.5

$S \rightarrow aBA \mid b$

$A \rightarrow SaB \mid Bb \mid \epsilon$

$B \rightarrow cB \mid Sd \mid \epsilon$

Solution:

First(S) = { a, b }

First(A) = { a, b, c, null }

First(B) = { c, a, b, null }

Follow(S) = { $, a, d }

Follow(A) = { $, a, d }

Follow(B) = { $, a, d, b }

Parse Table:

Non-Terminals	a	b	c	d	$
S	S→aBA Sync	S→b		Synce	Synce
A	~~S→aBA~~ A→ε	~~S→b~~	~~B→cB~~	A→ε	A→ε
B	~~S→aBA Sync~~	~~S→b~~	B→cB	B→ε	B→ε

(b)

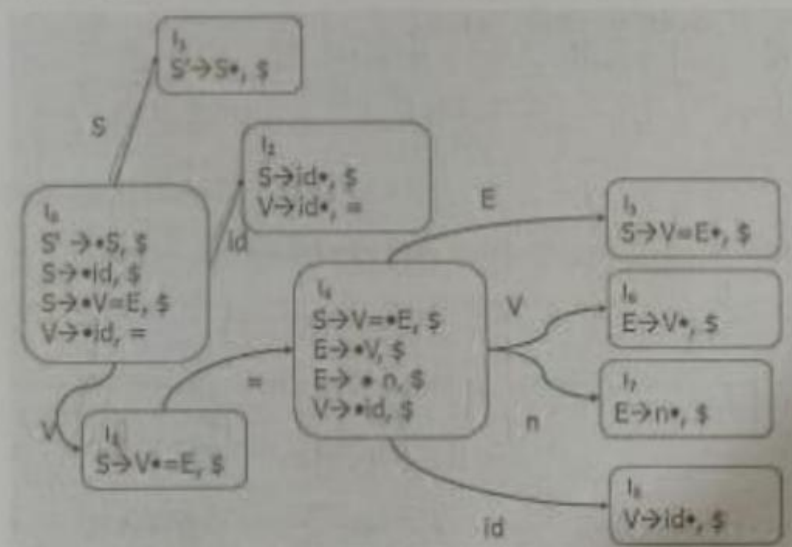
Only difference here is he gave us S only once (S → id | V =E) and same for E (E → V | n)

(b) Consider the following augmented grammar: *(10 points)*

10

S' → S
1) S → id
2) S → V = E
3) V → id
4) E → V
5) E → n

The Canonical LR(1) items for this grammar are given in the following transition diagram.



Construct LR(1) parse table for the above grammar using this transition diagram.

Solution:

STATE	ACTIONS				GOTO		
	Id	=	n	$	S	V	E
0	S2				1	3	
1				Accept			
2		R3		R1			

Q8

Consider the following assignment statement.

x = x * b – c + x * b

a. Construct abstract syntax tree (AST) for this statement.

Number of nodes in the AST:

b. Draw directed acyclic graph (DAG) for the above assignment statement.

Number of nodes in the DAG:

c. Represent the above assignment statement in three address intermediate code

$$b = (b + c) * c - d$$

d. Show the quadruple representation of the above three-address code.